

AI Engineering & Agentic AI Roadmap (2026)

SUMMARY

Current Position Assessment: Existing Knowledge includes Probability and Statistics, Data Analytics fundamentals, Machine Learning, Deep Learning, Basic Natural Language Processing (NLP), Python Programming, Java Programming, Backend Development (Node.js, Express), Databases (MongoDB), API Development, and Real-time Applications (Socket.IO). This places me beyond the beginner stage in AI and software development.

PROJECTS

PDF Chatbot | Project 1

- Upload PDF
- Ask Questions
- Generate Answers

Resume Analyzer | Project 2

- Resume Parsing
- Skill Extraction
- Feedback Generation

Meeting Summarizer | Project 3

- Summarization
- Action Items
- Follow-up Generation

College Notes Assistant | Allows querying personal notes.

Research Paper Assistant | Answers questions from uploaded papers.

Legal Document Assistant | Supports legal document retrieval.

Internship Finder Agent

- Search Internships
- Rank Opportunities
- Generate Reports

Research Agent

- Search Papers
- Summarize Findings
- Create References

Coding Assistant Agent

- Read Codebase
- Generate Code
- Debug Issues

HR Recruitment Workflow

- Resume → Screening → Ranking → Interview Scheduling

Customer Support Workflow

- Ticket → Classification → Response Generation → Escalation

Lead Generation Workflow

- Search → Data Extraction → Qualification → Outreach

RAG PDF Chatbot | Portfolio Projects
Resume Analyzer | Portfolio Projects
Research Paper Assistant | Portfolio Projects
AI Coding Assistant | Portfolio Projects
Internship Finder Agent | Portfolio Projects
Multi-Agent Research System | Portfolio Projects
AI Workflow Automation Platform | Portfolio Projects

TECHNICAL SKILLS

Probability and Statistics
Data Analytics fundamentals
Machine Learning
Deep Learning
Basic Natural Language Processing (NLP)
Python Programming
Java Programming
Backend Development (Node.js, Express)
Databases (MongoDB)
API Development
Real-time Applications (Socket.IO)
Building Generative AI applications
Developing AI agents
Creating Agentic AI workflows
Designing multi-agent systems
Deploying production-ready AI applications
LLM Fundamentals
Training Process
Fine-Tuning
Inference
Transformer Architecture
Self Attention
Multi Head Attention
Encoder
Decoder
Embeddings
Tokenization
Context Windows
Temperature
Top-K Sampling
Top-P Sampling
Prompt Engineering
Zero-Shot Prompting
One-Shot Prompting
Few-Shot Prompting
Chain of Thought Prompting
GPT
Gemini
Claude

Llama
Mistral
OpenAI API
Gemini API
Anthropic API
Function Calling
Tool Calling
Structured Outputs
JSON Mode
Streaming Responses
LangChain Basics
Prompt Templates
Chains
Output Parsers
Sentence Embeddings
Similarity Search
Chunking
Metadata Handling
Vector Search
Hybrid Search
Re-ranking
ChromaDB
FAISS
Pinecone
Agent Architecture
Planning
Reflection
Tool Usage
LangGraph
CrewAI
PydanticAI
Agno
MCP (Model Context Protocol)
Single Agent
Multi-Agent Systems
Supervisor Agent Pattern
Planner-Executor Pattern
n8n
Flowise
LangGraph Workflows
Triggers
Events
Conditional Logic
Human-in-the-loop Systems
FastAPI
Docker
Render
Railway

AWS Basics
Logging
Tracing
Evaluation
API Keys
Rate Limiting
User Authentication
Build LLM Applications
Build RAG Systems
Develop AI Agents
Create Multi-Agent Systems
Design AI Workflows
Use MCP
Deploy AI Applications
Integrate AI into Real Products

GENERATIVE AI VS AGENTIC AI

Generative AI focuses on creating content (Text generation, Image generation, Code generation, Audio generation, Video generation). Workflow: User Prompt → LLM → Generated Output. Characteristics: Reactive, One-step execution, No autonomous decision making, No long-term planning, Limited memory. Agentic AI extends Generative AI by allowing systems to plan, reason, use tools, and execute actions autonomously. Workflow: User Goal → Planning → Tool Selection → Reasoning → Action Execution → Feedback Loop → Final Result. Characteristics: Goal-oriented, Autonomous, Multi-step reasoning, Uses external tools, Maintains memory, Performs actions. Relationship: Generative AI = LLM. Agentic AI = LLM + Memory + Reasoning + Planning + Tool Usage + Autonomous Actions.

EVALUATION OF THE FREECODECAMP 30-HOUR GENERATIVE AI COURSE

Topics Covered: Introduction to Generative AI, OpenAI APIs, Gemini APIs, Hugging Face, LangChain, LLM Application Development, Vector Databases, Pinecone, ChromaDB, GPT Models, Gemini Models, Llama Models, Medical Chatbot, End-to-End LLM Projects. Relevance in 2026: Highly Relevant (LLM Fundamentals, OpenAI APIs, Gemini APIs, Hugging Face, Embeddings, Vector Databases, RAG Fundamentals), Moderately Relevant (LangChain Basics, Older OpenAI API Examples), Less Relevant (Older LangChain Syntax, Agentic AI Concepts). Rating: Learning Generative AI Fundamentals 8.5/10, Learning RAG 8/10, Learning AI Engineering 7.5/10, Learning Agentic AI 4/10. Conclusion: The course is worth completing because it provides a strong foundation in modern Generative AI systems.

ROADMAP PHASES AND DURATIONS

Phase 1: LLM and Generative AI Foundations (2–3 Weeks). Phase 2: AI Application Development (2–4 Weeks). Phase 3: Retrieval-Augmented Generation (RAG) (3–4 Weeks). Phase 4: Agentic AI (4–6 Weeks). Phase 5: AI Workflow Automation (2–4 Weeks). Phase 6: Deployment and Production (2–3 Weeks).

RESOURCES

FreeCodeCamp Generative AI Course; Hugging Face LLM Course; Andrew Ng Generative AI Courses.

FINAL RECOMMENDED LEARNING SEQUENCE

1. Complete the 30-Hour FreeCodeCamp Generative AI Course. 2. Build a PDF Chatbot. 3. Learn RAG thoroughly. 4. Build two RAG projects. 5. Learn LangGraph. 6. Learn Agent Design Patterns. 7. Build three AI Agents. 8. Learn MCP. 9. Learn n8n and Workflow Automation. 10. Build a Multi-Agent Production Project. 11. Deploy Everything. 12. Start applying for AI Engineering and Agentic AI opportunities.